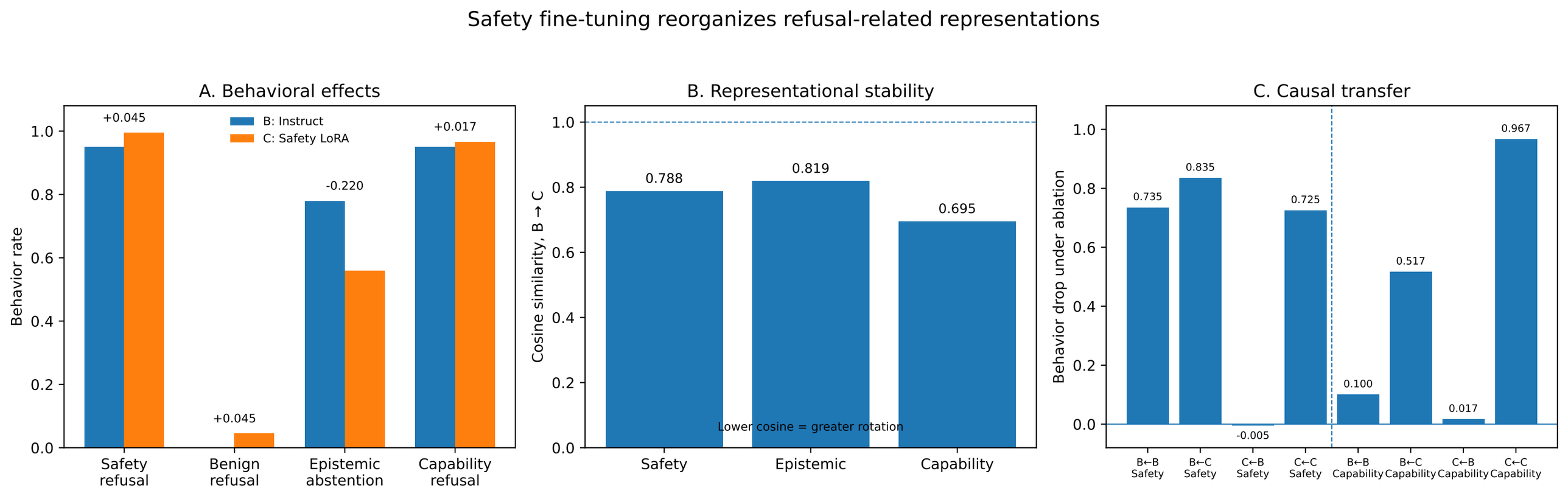
Safety fine-tuning reorganizes refusal-related representations
A. Behavioral effects
B: Instruct
C: Safety LoRA
Behavior rate
+0.045
+0.045
-0.220
+0.017
Safety refusal
Benign refusal
Epistemic abstention
Capability refusal
B. Representational stability
Cosine similarity, B → C
0.788
0.819
0.695
Lower cosine = greater rotation
Safety
Epistemic
Capability
C. Causal transfer
Behavior drop under ablation
0.735
0.835
-0.005
0.725
0.100
0.517
0.017
0.967
B←B Safety
B←C Safety
C←B Safety
C←C Safety
B←B Capability
B←C Capability
C←B Capability
C←C Capability